



GENERAL ZOOLOGY

COURSE (100Z)

(2018-2019)

Connective tissue

Connective

Origin: **tissues:**

All types of connective tissue originate from the mesoderm.

Types:

there are three different types of connective tissue:

- 1- C. T. proper** it may be loose (Areolar, Adipose, Reticular) or may be dense (Dense regular, Dense irregular and elastic).
- 2- Skeletal C. T.** (it constitutes both bone and cartilage)
- 3- Vascular C.T.** (Blood).

Components :

Typical connective tissue consists of:

- cells embedded in an extracellular matrix (ECM).
- ECM (Extracellular matrix) has a ground substance,
- fibers and tissue fluids.

Components of connective

- **Cells of connective tissue: tissues:**

- 1- **Fibrocytes**: this type of cells is present mainly in proper connective tissue.
- 2- **Chondrocytes and osteocytes**: are present mainly in cartilages and bones (skeletal C.T.)
- 3- **Blood cells**: which are circulating in the plasma of the vascular connective tissue.

- **Fibers of connective tissue**

- 1- **Collagen Fibers** (white fibers):

Composed of the protein “collagen”.

- 2- **Elastic Fibers** (Yellow fibers):

Composed of elastin protein, less strength than collagen.

- 3- **Reticular fibers**:

Fine collagenous fibers; branched and form networks; surround small blood vessels and support tissue of organs.

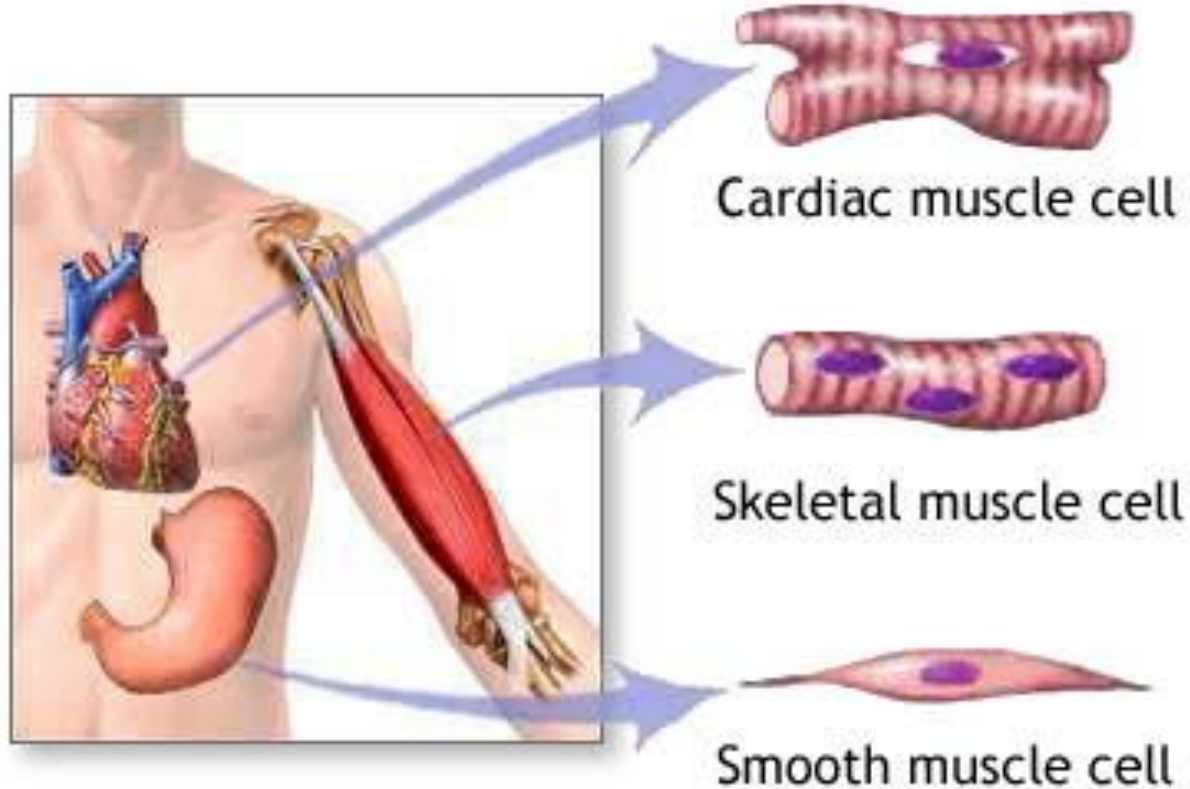
Functions of C. T.

1. Binding and support (ex. Tissues surrounding intestine, bone)
- 2 .Protection (ex. Blood)
- 3 .Thermal Insulation (Adipose C. T.)
4. Transportation (ex. Blood)

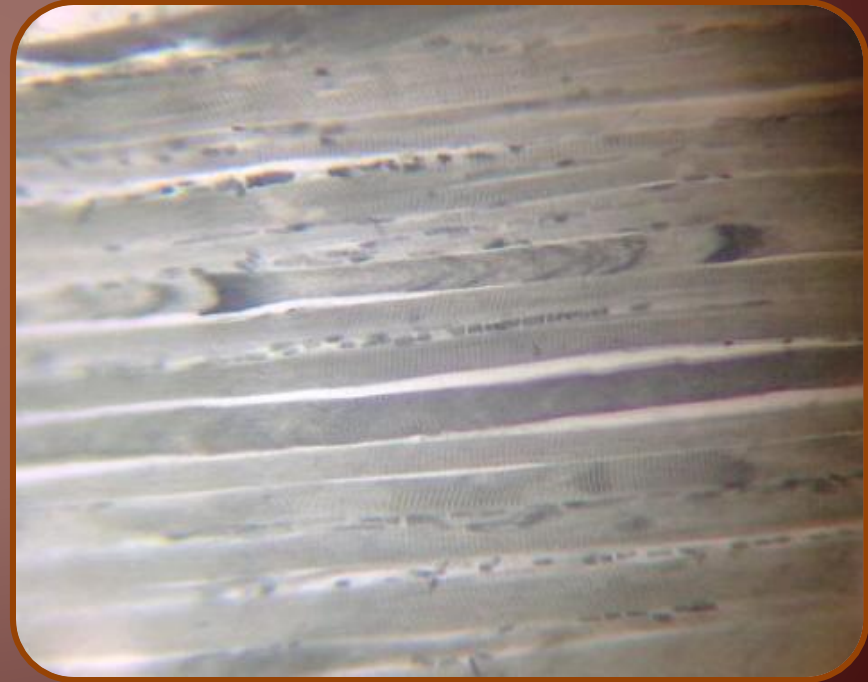
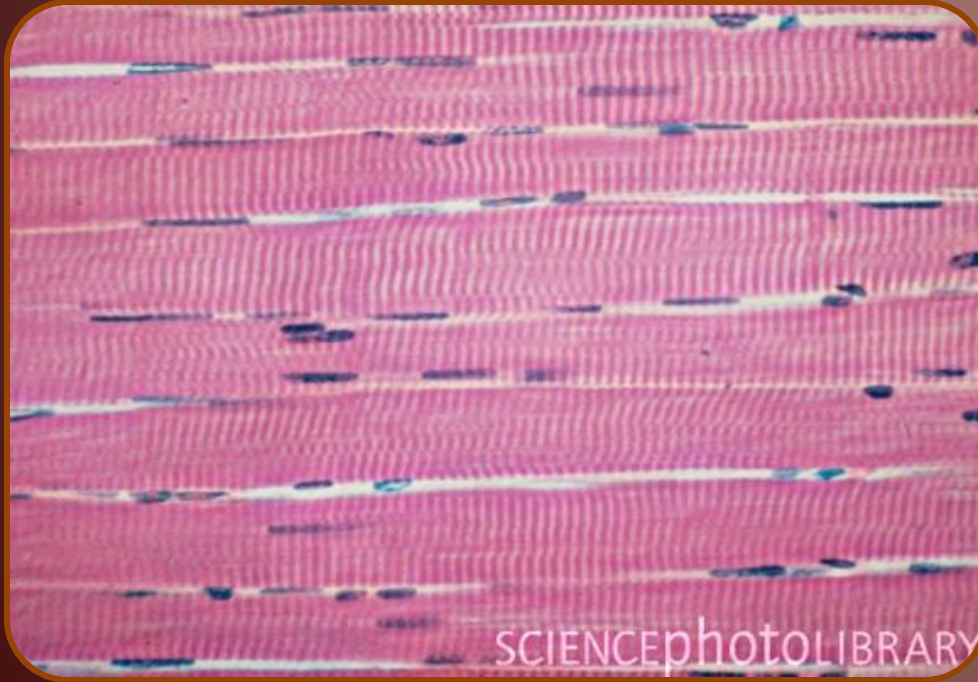
Muscular Tissue

Mostly animals has three types of muscles :

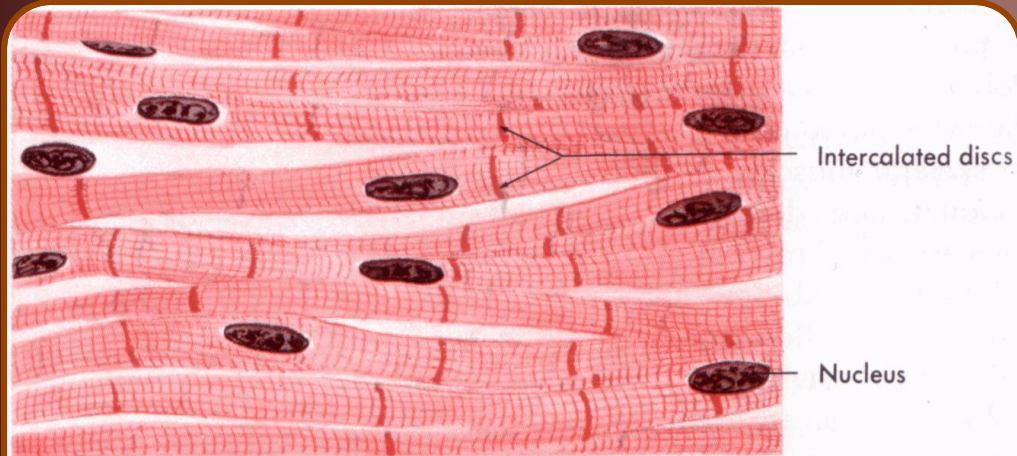
- Cardiac muscles - Smooth muscles - Skeletal muscles



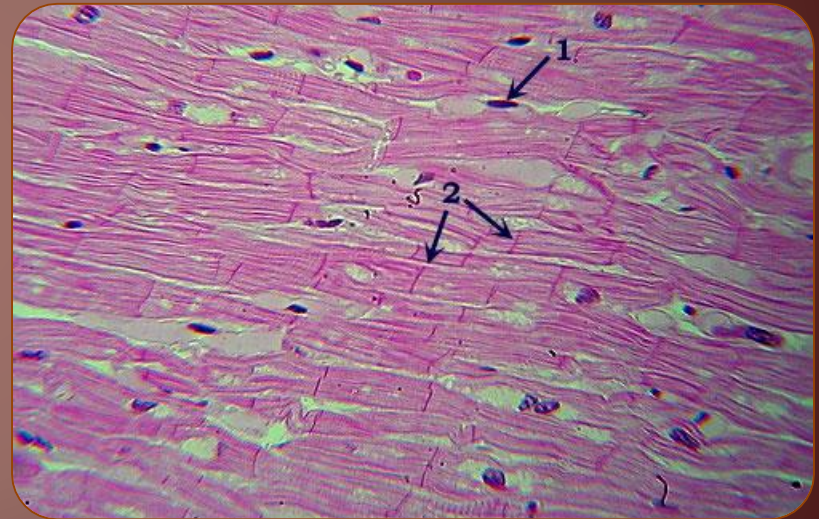
Skeletal Muscles



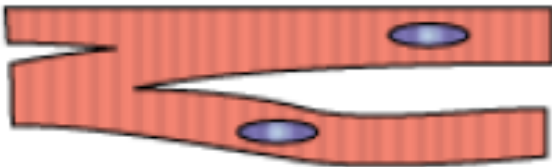
Cardiac Muscles



Cardiac or striated involuntary muscle tissue.



Cardiac or striated involuntary muscle tissue.

Skeletal		Voluntary	Striated	Multinucleated	Non-branched
Cardiac		Involuntary	Striated	Single nucleus	Branched
Smooth		Involuntary	Nonstriated	Single nucleus	Tapered

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Fig. Differences between different muscles